

Power Bank for Smartphones

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This power bank can be used for charging smartphones. The circuit uses two integrated modules and a lithium-ion battery. The first module is a lithium-ion battery charger and the second is a DC-DC boost converter module.

Circuit and working

Circuit diagram of the power bank is shown in Fig. 1. It consists of a USB lithium-ion charger module (USB-IN), 3.7V/2600mAh lithium-ion battery (BATT.1), on/off switch (S1) and DC-DC boost converter module (USB-OUT).

Here, a lithium-ion charger module based on TP4056 IC, rechargeable lithium-ion battery (Samsung type 18650) and DC-DC converter module based on pulse frequency modulation (PFM) technology are used.

The battery charger module, shown in Fig. 2, is designed around a dedicated lithium-ion battery charger TP4056 chip and populated

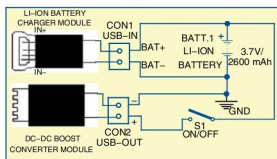


Fig. 1: Circuit diagram of the power bank



Fig. 2: TP4056 lithium-ion battery charger module



Fig. 3: DC-DC boost converter module

with SMD components. This on-board charge controller chip handles BATT.1 charging operation by processing the 5V DC input supply received through the USB socket (or through IN+ and IN- terminals). Output termi-

nals (BAT+ and BAT-) can be directly connected to BATT.1. Two onboard SMD LEDs located on top of the circuit board provide charging-status indications.

BATT.1 is used as the power reservoir. Since only 3.7V DC supply is available from BATT.1, DC-DC boost converter is used to cater to the stable 5V DC supply at output. If input voltage of 0.9V to 5V DC is available, this converter gives stable 5V DC output through its USB socket, with conversion efficiency up to 96 per cent.

The DC-DC boost converter module, shown in Fig. 3, is an SMD module with a PFM chip at its heart. Input DC supply fed through input terminals (+ and -) is processed by this dedicated chip to give a stable USB standard DC supply via the standard USB socket at its output. An onboard SMD LED, fitted near the input terminals, works as a power-status indicator. Switch S1 is included to route DC supply from BATT.1 to the converter.

Almost all smartphones look for signals/levels on D+ and D- of the USB interface that indicate the charger's current capability. A full-speed device will use a pull-up resistor attached to D+, as shown in Fig. 4, to satisfy itself as a full-speed device. The pull-up resistor at the device end will also be used by the host or web to detect the presence of a device connected to its port. Without this resistor logic, the USB assumes there is nothing connected to the bus.

In case of a charging error, pull down D+ line of the USB output socket (in converter module) with the help of a 200-kilo-ohm resistor as shown in Fig. 5.

Construction and testing

Assemble all components of the power bank circuit as per Fig. 1. Follow the testing operation given below:

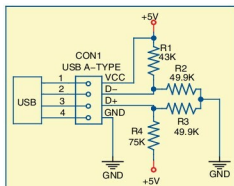


Fig. 4: Pull-up resistor configuration at D+ and D- pins

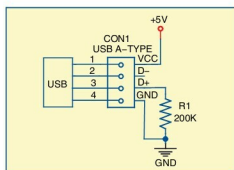


Fig. 5: Pull-down resistor configuration at D+ pin

1. Connect the charger module to CON1 (USB-IN) either through an AC adaptor or provide 5V DC supply through a PC or laptop using a USB cable for charging BATT.1.

2. Connect BATT.1 across CON1 for charging. A red-coloured LED will turn on, showing the charging status of BATT.1. When the battery is fully charged, a blue-coloured LED will turn on. You may remove the charger connected at USB-IN.

3. You can now charge your smartphone by connecting it to the USB socket of the converter module (USB-OUT) using a USB cable and closing switch S1. A red-coloured LED of the converter module will turn on, showing the charging status of the smartphone.

EFY note. Connect the terminals of BATT.1 to USB-IN and USB-OUT with correct polarity. ●



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